Lab 7 Rubric: Moment of Inertia

Out of 100 points. This is the rubric on the LB7-M5 assignment in Brightspace. Lab 7 page · Word copy · PDF copy

Use this rubric as your writing guide: the criteria follow the lab report template top to bottom. Full-credit descriptors say exactly what a complete answer contains, so read them before you write rather than after.

Grammar, spelling, and formatting problems are assessed 1 point per occurrence, capped at 3 points for the report. A fault already scored under a criterion is not deducted again as a formatting occurrence. Deadline: a report submitted by the deadline may be revised and resubmitted up to two more times. A report that arrives after the deadline but before grades are posted is graded as on time but cannot be revised; one that arrives after grades are posted is late, may take a penalty, and gets no detailed feedback.

The phrase "human error" earns no credit anywhere in the report. A percent difference under 10% between your experimental moment of inertia and $\frac{1}{2}M_D r_D^2$ may be attributed to ordinary experimental uncertainty, provided you say so explicitly and name a mechanism. Building the torque from $M_H g$ rather than $M_H(g - a)$ is a physics error, not a rounding one, and is marked as such.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Heading	3	**3** Names, course and section, lab title and date, all present and correctly formatted. No template placeholders.	**2** One item missing or misformatted.	**2** Template placeholders left in the heading block.	**1** Several items missing.	**0** No heading block.
Abstract	7	**7** Four sentences or fewer. States what was measured, gives I from the graph with its uncertainty and the percent difference against $\frac{1}{2}M_D r_D^2$, and states whether $\tau = I\alpha$ was verified. No methods.	**6** All content present but runs long or includes method detail.	**4** Reports a result but omits the uncertainty or the comparison.	**2** Describes the lab but reports no numbers.	**0** No abstract.
Introduction	8	**8** Derives $r_p[M_H(g - r_p \alpha)] = I\alpha$ line by line from the two diagrams and the constraint, with every symbol defined and the two radii explicitly distinguished. States that the slope of α against τ is 1/I. Ends with a testable hypothesis.	**7** Derivation present but a step or the radius distinction is thin.	**5** The relation is quoted rather than derived, or T is taken as $M_H g$.	**3** Equation given with no derivation.	**0** No introduction.
Experimental: Materials & Setup	8	**8** Apparatus in prose; side-on AND top-down sketches with BOTH radii marked and distinguished; free-body and torque diagrams; pulley radius, disc radius and disc mass all given.	**7** Sketches and diagrams present; one radius unlabelled.	**5** Only one sketch, or the two radii not distinguished.	**3** Bare equipment list, or no diagrams.	**0** No materials or setup description.
Experimental: Procedure	8	**8** Written in own words as prose, reproducible. Says how α was recorded, what was varied, and notes the no-slip requirement.	**7** Reproducible and in own words; the no-slip point is absent.	**5** Reproducible but vague about what was varied.	**3** Copied or closely paraphrased from the handout.	**0** No procedure.
Results: Data Tables	12	**12** Table of all six runs carrying M_H, α, a, $M_H g$, $M_H a$, T and τ, so the widening gap between $M_H g$ and T is visible. Captions above, units in header rows.	**10** Table present; minor issues with captions, units or significant figures.	**7** Table present but $M_H g$ and $M_H a$ are not shown separately, hiding the tension correction.	**4** Fewer than six runs tabulated, or torque omitted.	**0** No data tables.
Results: Graphs	8	**8** α against τ with a fitted straight line, axes labelled with units, caption below. The slope is identified as 1/I.	**7** Graph present and correctly plotted; caption or labels incomplete.	**5** Axes transposed, so the slope is I rather than 1/I, without comment.	**3** Points joined dot-to-dot, or no fitted line.	**0** No graph.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Results: Sample Calculations	**13**	**13** One worked sample of each type with units at every step: a from α, tension, torque, I from the slope, δI, and percent difference. Torque in N·m and said not to be joules.	**11** All types present with minor unit or rounding slips.	**8** Three or four of the six types worked.	**4** Some types missing, or units only in the final answer.	**0** No sample calculations.
Discussion	**15**	**15** Opens with I from the slope and its percent difference from ½M_D r_D², then argues it. Chooses between the two values with a reason, addresses the intercept, accounts for the DIRECTION of the discrepancy, and tests a physical explanation against the data.	**12** Most of the above; one element thin, usually the direction of the discrepancy.	**9** Interprets the result but does not engage with what the model leaves out.	**5** Restates the results without interpreting them.	**0** No discussion.
Additional Questions	**8**	**8** Three of the five chosen and named, each answered fully in a dedicated section, in paragraphs with working shown and not restating the question.	**6** Three answered but one thinly, or the three chosen are not named.	**4** Answered inside another section, or only two answered.	**2** Answered as a bare list, or fewer than two answered.	**0** Not answered.
Conclusion	**5**	**5** Three to four sentences: numerical result, definitive verdict, one concrete improvement.	**4** Present but missing the numbers or the verdict.	**3** Present with numbers but no verdict.	**2** Vague, or a repeat of the discussion.	**0** No conclusion.
References	**5**	**5** At least one source beyond the textbook; the lab-analysis note and the handout count. Author, title and date, consistently formatted.	**4** Multiple references but formatting incomplete.	**3** Textbook and handout only, properly formatted.	**2** Handout only.	**0** No references.
Total	**100**					

Overall: Complete 90+ · Nearly there 75+ · Needs work 50+ · Missing below 50.

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